HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - III

Probability Theory

Topics covered in this unit :

- Probability, probability axioms
- Addition law and multiplication law of probability
- Conditional probability, Bayes' theorem
- Random variables (discrete and continuous)
- Probability density functions and their properties
- Mathematical expectation

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Basic Concepts of Probability

Term	Meaning
Random experiment	Outcome cannot be predicted with certainty, but all possible outcomes are known (tossing a coin)
Sample space S	Set of ALL possible outcomes (die: S = {1,2,3,4,5,6})
Event E	A subset of S (getting an even number: {2,4,6})
Simple / compound event	Single outcome / combination of outcomes
Sure and impossible event	S itself / empty set (null)
Mutually exclusive	Cannot occur together: A ∩ B = empty
Exhaustive events	Together they cover S: A1 ∪ A2 ∪ ... = S
Independent events	Occurrence of one does not affect the other
Equally likely	Each outcome has the same chance

1.1 Definitions of probability

Classical: P(E) = (number of favourable cases) / (total number of equally likely cases) = m / n

- Statistical (relative frequency): P(E) = lim (n -> inf) (m/n), where m = number of times E occurs in n trials.
- Axiomatic (Kolmogorov): see below.

1.2 Axioms of probability

Kolmogorov's axioms

- Axiom 1: 0 <= P(E) <= 1 for every event E (non-negativity)
- Axiom 2: P(S) = 1 (certainty)
- Axiom 3: If E1, E2, ... are mutually exclusive, then P(E1 ∪ E2 ∪ ...) = P(E1) + P(E2) + ... (additivity)

1.3 Consequences

Result	Formula
Impossible event	P(empty) = 0
Complement	P(A') = 1 - P(A)
Subset	If A is a subset of B then P(A) <= P(B)
Difference	P(A - B) = P(A) - P(A n B)
Neither A nor B	P(A' n B') = 1 - P(A U B)
Exactly one of A, B	P(A) + P(B) - 2 P(A n B)

Example 1

Two dice are thrown. Find P(sum = 8). Total outcomes = 36. Favourable: (2,6),(3,5),(4,4),(5,3),(6,2) = 5. P = 5/36 .

Example 2

A card is drawn from a pack of 52. P(King or heart)? Kings = 4, hearts = 13, King of hearts counted twice: (4 + 13 - 1)/52 = 16/52 = 4/13 .

Example 3

From a box with 4 red and 6 black balls, 3 are drawn. P(all red) = C(4,3)/C(10,3) = 4/120 = 1/30 . P(2 red, 1 black) = C(4,2) C(6,1)/120 = 36/120 = 3/10 .

2. Laws of Probability

2.1 Addition law

General (any two events)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) \text{ if } A, B \text{ are mutually exclusive}$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

2.2 Conditional probability

Probability of A given that B has occurred :

$$P(A \mid B) = P(A \cap B) / P(B), \quad P(B) > 0$$

2.3 Multiplication law

General

$$P(A \cap B) = P(A)\, P(B \mid A) = P(B)\, P(A \mid B)$$

Independent events

$$P(A \cap B) = P(A)\, P(B) \text{ if } A, B \text{ are INDEPENDENT}$$

$$P(A \cap B \cap C) = P(A)\, P(B|A)\, P(C \mid A \cap B)$$

Independent vs mutually exclusive

- Mutually exclusive: $A \cap B$ = empty $\rightarrow P(A \cap B) = 0$ (cannot happen together)
- Independent: $P(A \cap B) = P(A)\, P(B)$
- Two events with non-zero probabilities that are mutually exclusive are never independent
- If A and B are independent, so are A and B', A' and B, A' and B'

Example 1

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Then $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$.

$P(A|B) = 0.2/0.4 = 0.5 = P(A) \rightarrow$ A and B are independent.

Example 2 - drawing without replacement

Two cards are drawn without replacement from 52. P(both aces) = (4/52)(3/51) = 1/221 .

Example 3

A can solve a problem with probability 1/2, B with 1/3. If both try independently, P(problem solved) = 1 - P(neither) = 1 - (1/2)(2/3) = 2/3 .

Example 4

A bag has 5 white, 3 black balls. Two drawn one by one with no replacement. P(first white, second black) = (5/8)(3/7) = 15/56 .

3. Total Probability and Bayes' Theorem

3.1 Theorem of total probability

If B1, B2, ..., Bn are mutually exclusive and exhaustive events (a partition of S) with P(Bi) > 0, then for any event A:

$$P(A) = \sum_{i=1}^{n} P(B_i)\,P(A \mid B_i)$$

3.2 Bayes' theorem

Bayes' theorem

- $P(B_K \mid A) = P(B_K)\,P(A \mid B_K) \,/\, \sum_{i=1}^{n} P(B_i)\,P(A \mid B_i)$
- P(Bi) = prior probabilities (before the event), P(A | Bi) = likelihood ; P(BK | A) = posterior probability (after A is known)
- Used to find the probability of a cause given its effect

Tree method

Choose source Bi with P(Bi) → Event A occurs with P(A | Bi) → Add all branches for P(A); posterior = branch / total

Example 1 - Three machines

Machines M1, M2, M3 produce 50 %, 30 %, 20 % of items with defective rates 3 %, 4 %, 5 %. An item picked at random is defective. Find P(it came from M3).

Machine	P(Bi)	P(D | Bi)	Product
M1	0.50	0.03	0.015
M2	0.30	0.04	0.012
M3	0.20	0.05	0.010
Total	1.00		0.037

P(D) = 0.037. P(M3 | D) = 0.010/0.037 = 0.2703 (27 %).

Example 2 - Medical test

1 % of people have a disease. Test is 95 % positive for diseased and 2 % false positive for healthy. P(disease | positive) = (0.01 × 0.95)/(0.01 × 0.95 + 0.99 × 0.02) = 0.0095/0.0293 = 0.324 . Even a positive test gives only 32 % chance - because the disease is rare.

Example 3 - Urns

Urn I has 3 white, 2 black balls; Urn II has 2 white, 4 black. An urn is chosen at random and a ball is drawn - it is white. P(from Urn I) = (1/2)(3/5)/[(1/2)(3/5) + (1/2)(2/6)] = 0.3/(0.3 + 0.1667) = 0.643 .

4. Random Variables

A random variable (r.v.) X is a function that assigns a real number to each outcome of the sample space: $X : S \to R$. Written in capital letters (X), its values in small letters (x).

Discrete r.v.	Continuous r.v.
Takes countable values (finite or infinite list)	Takes any value in an interval (uncountable)
Number of heads, defects, customers	Height, weight, time, temperature

Discrete r.v.	Continuous r.v.
Described by probability mass function (p.m.f.) $p(x)$	Described by probability density function (p.d.f.) $f(x)$
$P(X = x) = p(x)$ can be > 0	$P(X = x) = 0$ for any single point

4.1 Discrete: probability mass function

p.m.f. conditions

- $p(x_i) = P(X = x_i) \geq 0$ for all i
- Sum of $p(x_i) = 1$
- Distribution function (c.d.f.): $F(x) = P(X \leq x)$ = Sum of $p(x_i)$ for $x_i \leq x$

Example

Two coins are tossed; X = number of heads. $S = \{HH, HT, TH, TT\}$.

x	0	1	2
$p(x)$	1/4	1/2	1/4
$F(x)$	1/4	3/4	1

Check: $1/4 + 1/2 + 1/4 = 1$. $P(X \geq 1) = 3/4$.

Example - find K

$p(x) = K\,x$ for $x = 1, 2, 3, 4, 5$. Sum $= K(1+2+3+4+5) = 15K = 1 \rightarrow K = 1/15$.

$P(X \leq 2) = 3/15 = 1/5$.

4.2 Continuous: probability density function

p.d.f. $f(x)$ properties

- $f(x) \geq 0$ for all x
- $\int_{-\infty}^{\infty} f(x)\,dx = 1$ (total area = 1)
- $P(a \leq X \leq b) = \int_a^b f(x)\,dx$ = area under the curve (end points do not matter)
- c.d.f.: $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$; $f(x) = F'(x)$
- F is non-decreasing, $F(-\infty) = 0$, $F(\infty) = 1$, $0 \leq F \leq 1$

Example 1

f(x) = K x^2 for 0 <= x <= 3, 0 elsewhere. Int (0 to 3) K x^2 dx = 9 K = 1 -> K = 1/9 . P(1 <= X <= 2) = (1/9)(8 - 1)/3 = 7/27 . F(x) = x^3/27 (0 <= x <= 3).

Example 2

f(x) = 2 e^(-2x), x >= 0 (exponential). P(X > 1) = Int (1 to inf) 2e^(-2x) dx = e^(-2) = 0.1353 . F(x) = 1 - e^(-2x).

Example 3

f(x) = K(1 - x^2), -1 <= x <= 1: Int = K [x - x^3/3] from -1 to 1 = K(4/3) = 1 -> K = 3/4 .

5. Mathematical Expectation

5.1 Definition

Mean = expected value = mu

Discrete: E(X) = Sum xi p(xi) Continuous: E(X) = Int (-inf to inf) x f(x) dx

Expectation of a function

E[g(X)] = Sum g(xi) p(xi) or Int g(x) f(x) dx

5.2 Variance and standard deviation

Var(X) = E[(X - mu)^2] = E(X^2) - [E(X)]^2 , sigma = sqrt(Var(X))

5.3 Properties of expectation and variance

Expectation	Variance
E(c) = c	Var(c) = 0
E(aX + b) = a E(X) + b	Var(aX + b) = a^2 Var(X)
E(X + Y) = E(X) + E(Y) (always)	Var(X + Y) = Var X + Var Y if independent
E(XY) = E(X)E(Y) if independent	Var(X - Y) = Var X + Var Y if independent

Expectation	Variance
E(X^2) >= [E(X)]^2	Var(X) >= 0

Moments

r-th moment about origin: mu'_r = E(X^r). r-th central moment: mu_r = E[(X - mu)^r]. mu_1 = 0, mu_2 = Var(X). Moment generating function: M(t) = E(e^(tX)); mu'_r = d^r M/dt^r at t = 0.

Example 1 - discrete

X = number of heads in 2 tosses: E(X) = 0(1/4) + 1(1/2) + 2(1/4) = 1 . E(X^2) = 0 + 1/2 + 4/4 = 3/2. Var(X) = 3/2 - 1 = 1/2 .

Example 2 - die

X = score on a die. E(X) = (1 + 2 + ... + 6)/6 = 3.5 . E(X^2) = 91/6. Var = 91/6 - 49/4 = 35/12 = 2.917 .

Example 3 - continuous

f(x) = x/2 for 0 <= x <= 2. Check: Int x/2 dx = 1 OK. E(X) = Int x (x/2) dx = [x^3/6] 0 to 2 = 4/3 . E(X^2) = Int x^3/2 dx = [x^4/8] = 2. Var = 2 - 16/9 = 2/9 .

Example 4 - gain in a game

You win Rs.100 if a die shows 6, lose Rs.20 otherwise. E(gain) = 100(1/6) - 20(5/6) = 16.67 - 16.67 = 0 -> fair game.

Example 5 - E(2X + 3)

If E(X) = 4 and Var(X) = 9: E(2X + 3) = 11; Var(2X + 3) = 4 x 9 = 36 .

6. More Results on Expectation

6.1 Moment generating function (M.G.F.)

$$M_X(t) = E(e^{tX}) = \sum e^{tx} p(x) \text{ or } \int e^{tx} f(x)\,dx$$

- Moments: mu'_r = (d^r / dt^r) M_X(t) at t = 0 ; so mean = M'(0), E(X^2) = M''(0).

- M.G.F. of aX + b is e^(bt) M_X(a t); for independent X, Y: M_(X+Y)(t) = M_X(t) M_Y(t).
- M.G.F. uniquely determines the distribution.

Example

p(x) = (1/2)^x, x = 1, 2, 3, ... Then M(t) = Sum (e^t/2)^x = (e^t/2)/(1 - e^t/2) = e^t/(2 - e^t). M'(0) = 2 -> E(X) = 2 .

6.2 Chebyshev's inequality

P(|X - mu| >= K sigma) <= 1/K^2 or P(|X - mu| < K sigma) >= 1 - 1/K^2

Valid for any distribution with finite mean and variance. Example: for K = 2, at least 75 % of values lie within mu +/- 2 sigma; for K = 3, at least 88.9 %.

Example

A r.v. has mean 12 and variance 9. P(6 < X < 18) >= ? Here K sigma = 6, sigma = 3 -> K = 2. Bound = 1 - 1/4 = 0.75 .

6.3 Median, mode (continuous)

- Median M: Int (-inf to M) f(x) dx = 1/2. Mode: value where f(x) is maximum (f' = 0, f'' < 0).
- Example: f(x) = 3x^2 on (0,1): median: M^3 = 1/2 -> M = 0.794; mode = 1 (f increasing).

7. Practice - Worked Problems

Problem 1 - Conditional probability

In a class 60 % are boys, 30 % of boys and 20 % of girls play cricket. A student plays cricket. P(boy)? P(play) = 0.6(0.3) + 0.4(0.2) = 0.18 + 0.08 = 0.26. P(boy | play) = 0.18/0.26 = 0.692 .

Problem 2 - at least one

A coin is tossed 4 times. P(at least one head) = 1 - (1/2)^4 = 15/16 .

Problem 3 - Independent events

P(A) = 1/3, P(B) = 1/4, independent. P(A ∪ B) = 1/3 + 1/4 - 1/12 = 1/2 . P(A' ∩ B') = (2/3)(3/4) = 1/2 (= 1 - P(A ∪ B)).

Problem 4 - Bayes with three boxes

Box 1: 2 red 3 blue, Box 2: 3 red 2 blue, Box 3: 4 red 1 blue. A box is chosen randomly and a red ball is drawn. P(Box 3)? P(R) = (1/3)(2/5 + 3/5 + 4/5) = 3/5. P(B3 | R) = (1/3)(4/5)/(3/5) = 4/9 .

Problem 5 - Spam filter

30 % of mails are spam. The word 'offer' appears in 80 % of spam and 10 % of non-spam. P(spam | offer) = 0.3 x 0.8/(0.3 x 0.8 + 0.7 x 0.1) = 0.24/0.31 = 0.774 .

Problem 6 - Discrete distribution

x	-2	-1	0	1	2
p(x)	0.1	0.2	0.3	0.3	0.1

E(X) = -0.2 - 0.2 + 0 + 0.3 + 0.2 = 0.1 . E(X^2) = 0.4 + 0.2 + 0 + 0.3 + 0.4 = 1.3. Var = 1.3 - 0.01 = 1.29 .

Problem 7 - Continuous with c.d.f.

f(x) = 6x(1 - x), 0 < x < 1. Check: 6[x^2/2 - x^3/3] = 1. E(X) = ∫ 6x^2(1 - x) dx = 6(1/3 - 1/4) = 1/2 . E(X^2) = 6(1/4 - 1/5) = 3/10. Var = 0.3 - 0.25 = 0.05 .

Problem 8 - Find K and probability

f(x) = K e^(-3x), x > 0. ∫ = K/3 = 1 -> K = 3. P(X < 1) = 1 - e^(-3) = 0.9502 . Mean = 1/3.

Problem 9 - Counting and probability

A committee of 3 is formed from 5 men and 4 women. P(exactly 2 women) = C(4,2) C(5,1)/C(9,3) = 30/84 = 5/14 .

Problem 10 - Birthday-type

P(all 3 people have different birthdays) = (365/365)(364/365)(363/365) = 0.9918 ; so P(at least two share) = 0.0082.

Key formula	Use
nPr = n!/(n-r)!, nCr = n!/(r!(n-r)!)	Counting favourable cases
P(A|B) = P(A∩B)/P(B)	Conditional probability
Total probability + Bayes	Cause and effect problems
E(X), Var(X), M(t)	Mean, spread, moments

Quick revision - Unit III

- Axioms: 0 <= P <= 1, P(S) = 1, additivity; P(A ∪ B) = P(A) + P(B) - P(A ∩ B)
- P(A|B) = P(A ∩ B)/P(B); independent: P(A ∩ B) = P(A)P(B)
- Total probability P(A) = Sum P(Bi)P(A|Bi); Bayes: P(Bk|A) = P(Bk)P(A|Bk)/P(A)
- p.m.f.: Sum p = 1; p.d.f.: Int f dx = 1, f >= 0, f = F'
- E(X) = Sum x p or Int x f dx; Var = E(X^2) - [E(X)]^2